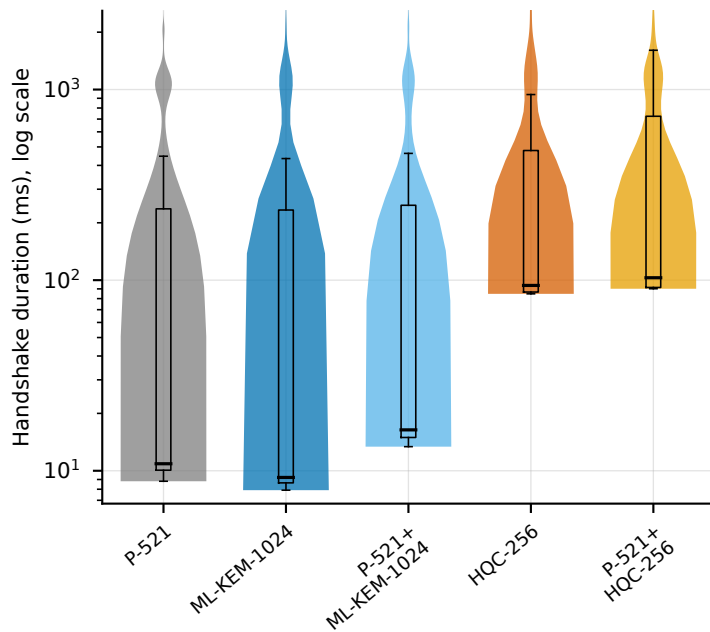
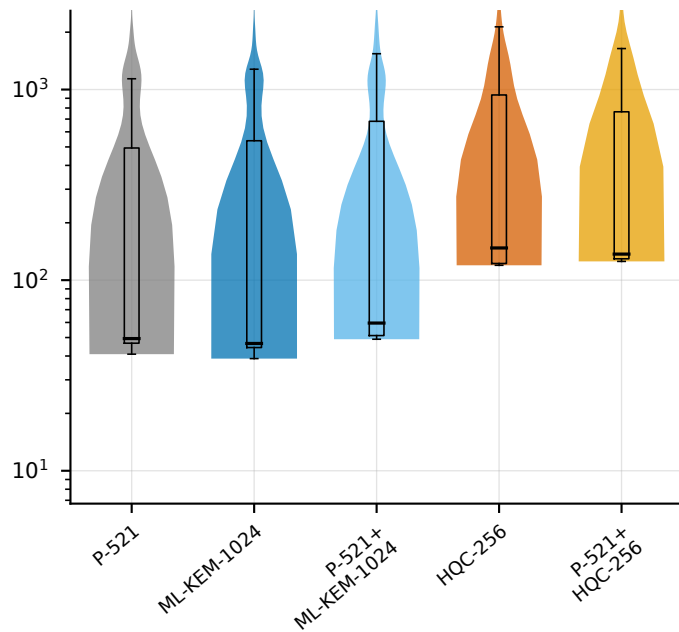


**(a)**

secp521r1 (classical)

**(b)**

mlDSA87 (post-quantum)

**(c)**

Classical KEX
  ML-KEM (pure)
  ML-KEM (hybrid)
  HQC (pure)
  HQC (hybrid)

| KEM               | secp521r1<br>Mean (ms) | secp521r1<br>CV (%) | secp521r1<br>Max (ms) | mlDSA87<br>Mean (ms) | mlDSA87<br>CV (%) | mlDSA87<br>Max (ms) |
|-------------------|------------------------|---------------------|-----------------------|----------------------|-------------------|---------------------|
| P-521             | 301.1                  | 196.1               | 4.15k                 | 491.6                | 181.0             | 7.66k               |
| ML-KEM-1024       | 302.4                  | 275.0               | 12.89k                | 431.6                | 173.8             | 9.76k               |
| P-521+ML-KEM-1024 | 313.3                  | 223.7               | 6.41k                 | 475.5                | 167.0             | 6.66k               |
| HQC-256           | 484.1                  | 175.4               | 11.34k                | 691.3                | 192.3             | 15.42k              |
| P-521+HQC-256     | 520.2                  | 161.1               | 8.71k                 | 651.1                | 234.4             | 26.67k              |